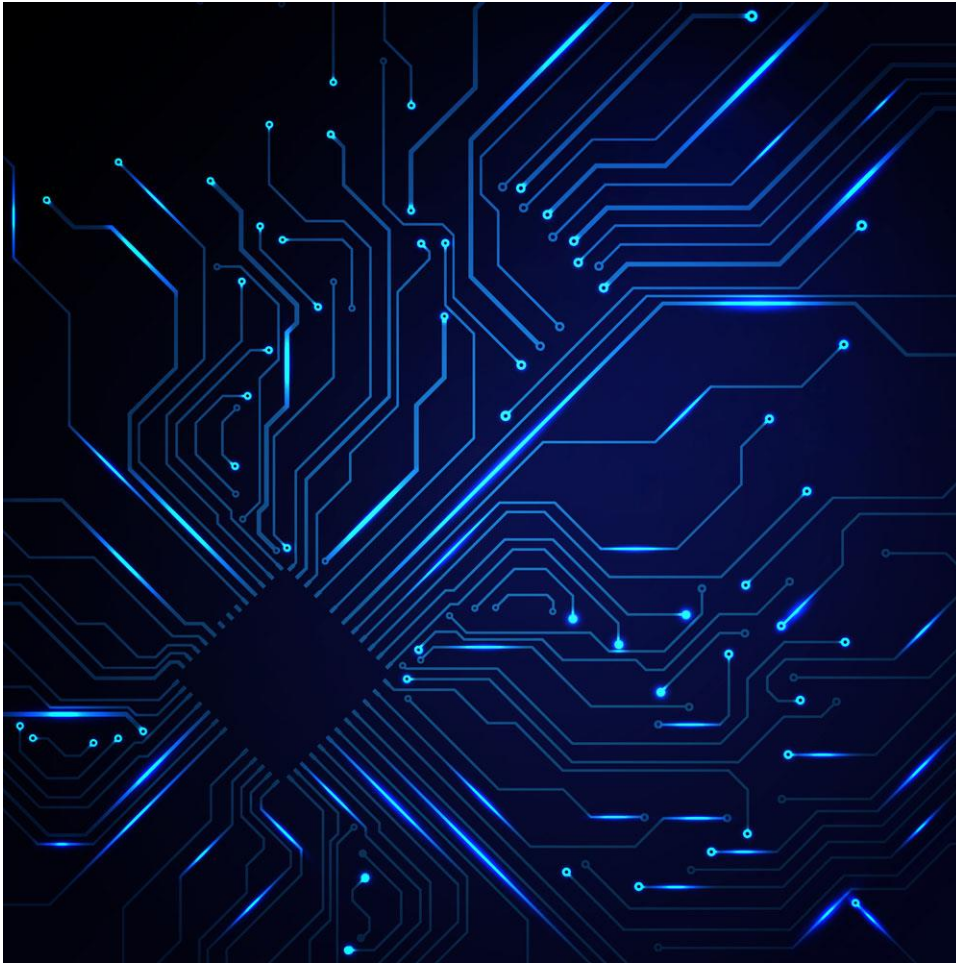




LAB 5 ADDER

Group : 14

Name : 黃翊宏

Student ID : 01357101

Note

1. All files must be converted to PDF format before submission.
2. **Submission deadline:**
Next Tuesday at 1:00 PM
3. Each person submits one assignment.

Lab 5-1

Truth Table (Voltage)

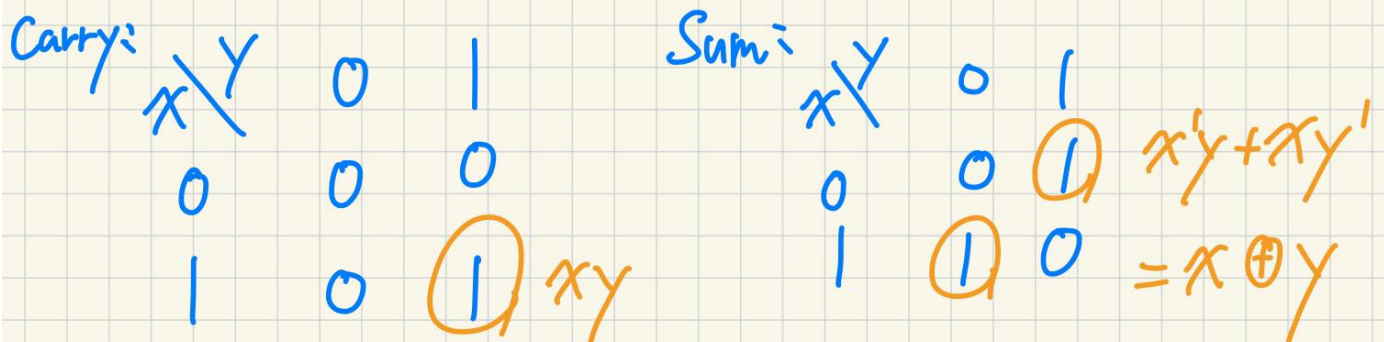
input		output	
A	B	C_{out}	S
0	0	0.56	0.13
0	1	0.56	2.16
1	0	0.56	2.16
1	1	2.24	0.13

$$C_{out} = xy$$

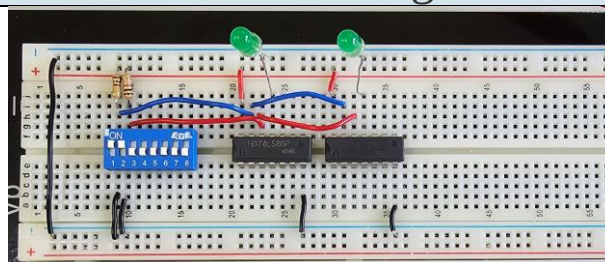
Simplify the results

$$S = x'y + xy' = x \oplus y$$

Karnaugh map



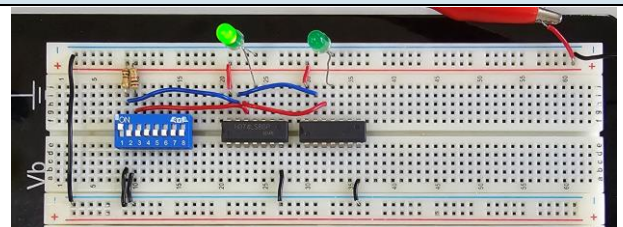
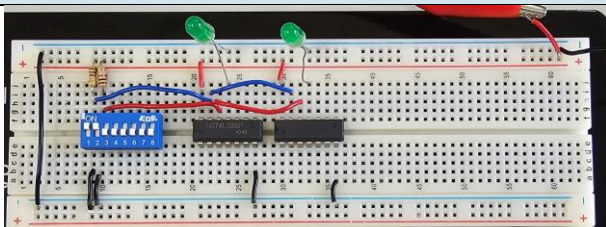
Breadboard Wiring Photo



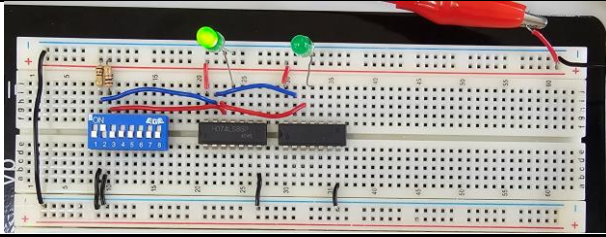
Breadboard Wiring Photos
(The output must be connected to LEDs.)

A = 0, B = 0

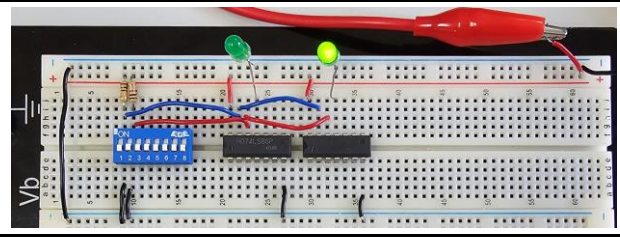
A = 0, B = 1



$A = 1, B = 0$



$A = 1, B = 1$



☆ Lab 5-2

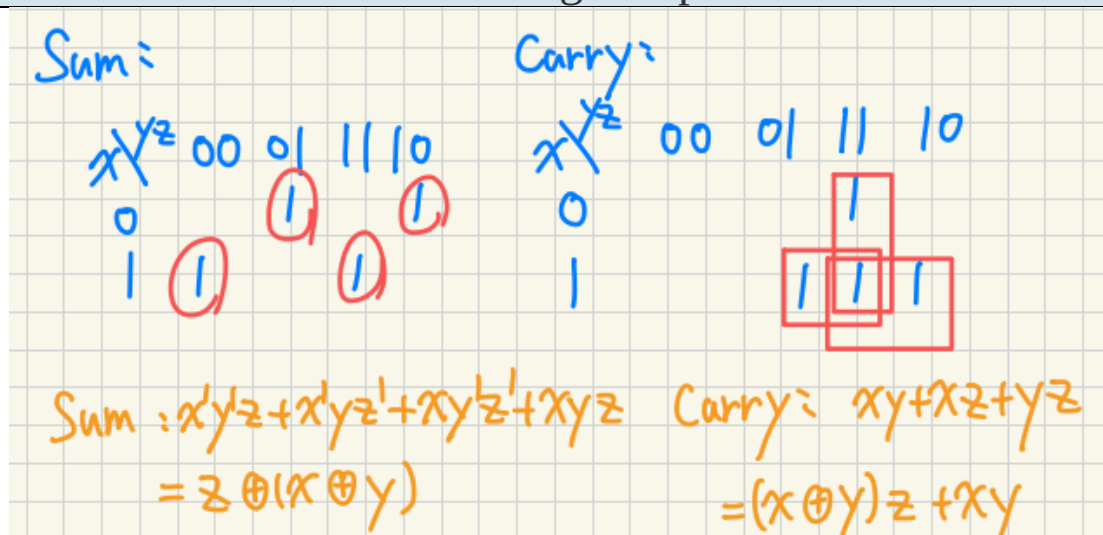
Truth Table (Voltage)				
input			output	
A	B	C_{in}	C_{out}	S
0	0	0	0.56	0.13
0	0	1	0.56	2.16
0	1	0	0.56	2.16
0	1	1	2.24	0.13
1	0	0	0.56	2.16
1	0	1	2.24	0.13
1	1	0	2.24	0.13
1	1	1	2.24	2.16

Simplify the results

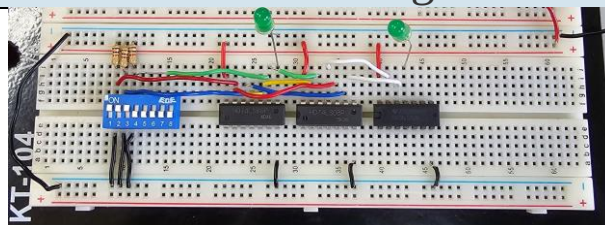
$$C_{out} = xy + xz + yz = (x \oplus y)z + xy$$

$$S = x'y'z + x'yz' + xy'z' + xyz = z \oplus (x \oplus y)$$

Karnaugh map

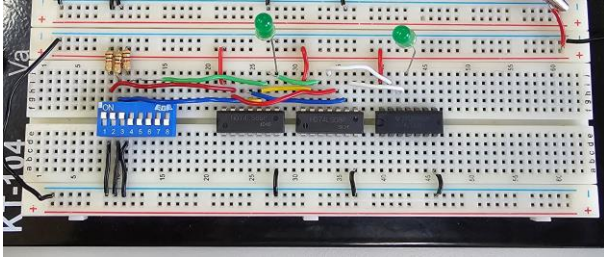


Breadboard Wiring Photo

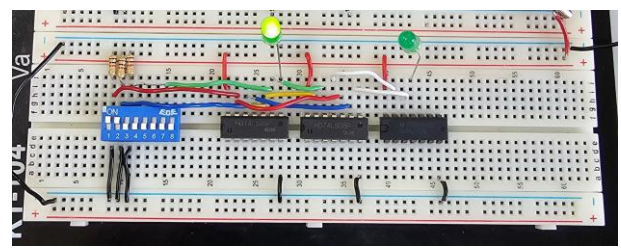


Breadboard Wiring Photos (The output must be connected to LEDs.)

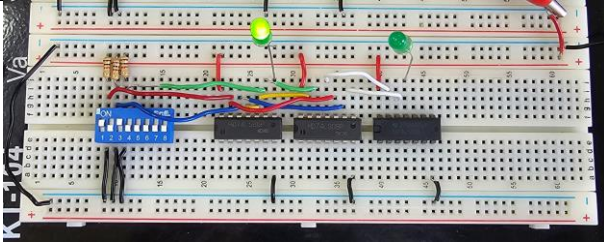
$A = 0, B = 0, C_{in} = 0$



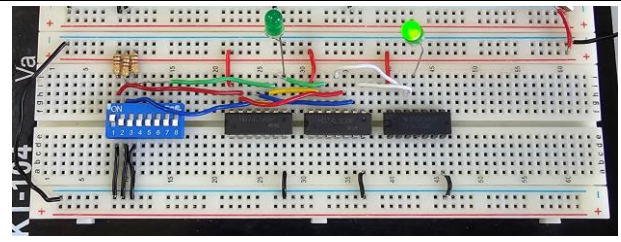
$A = 0, B = 0, C_{in} = 1$



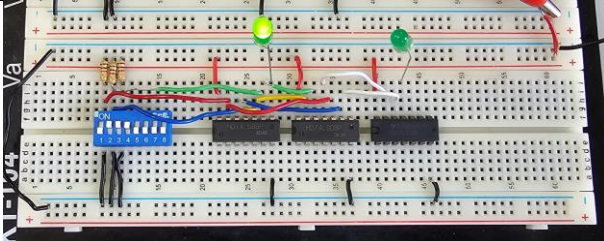
$A = 0, B = 1, C_{in} = 0$



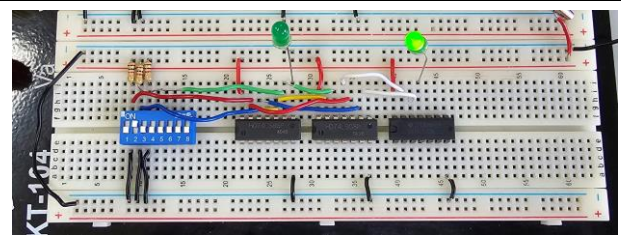
$A = 0, B = 1, C_{in} = 1$



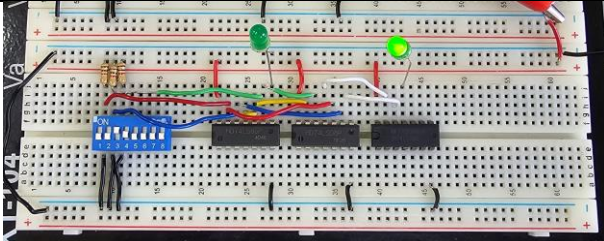
$A = 1, B = 0, C_{in} = 0$



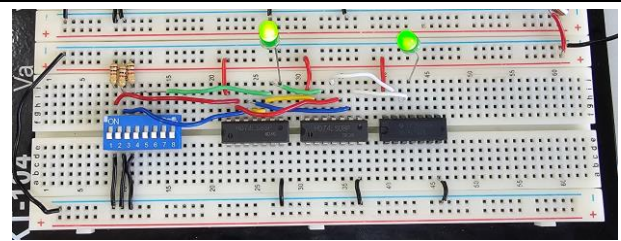
$A = 1, B = 0, C_{in} = 1$



$A = 1, B = 1, C_{in} = 0$



$A = 1, B = 1, C_{in} = 1$



★ Lab 5-3

Truth Table (Voltage)				
input			output	
A_1A_0	B_1B_0	C_0	S_1S_0	C_{out}
00	00	0.52	0.56 / 0.31	0.13
00	01	0.52	0.56 / 2.22	0.13
00	10	0.52	2.24 / 0.31	0.13
00	11	0.52	2.24 / 2.22	0.13
01	00	0.52	0.56 / 2.22	0.13
01	01	2.26	2.24 / 0.31	0.13
01	10	0.52	2.24 / 2.22	0.13
01	11	2.26	0.56 / 0.31	2.16
10	00	0.52	2.24 / 0.31	0.13
10	01	0.52	2.24 / 2.22	0.13
10	10	0.52	0.56 / 0.31	2.16
10	11	0.52	0.56 / 2.22	2.16
11	00	0.52	2.24 / 2.22	0.13
11	01	2.26	0.56 / 0.31	2.16
11	10	0.52	0.56 / 2.22	2.16
11	11	2.26	2.24 / 0.31	2.16
Simplify the results	$C_0 = A_0B_0$			
	$S_0 = A_0'B_0 + A_0B_0' = A_0 \oplus B_0$			
	$S_1 = A_1B_1'C_0' + A_1'B_1'C_0 + A_1B_1C_0 + A_1'B_1C_0'$			
	$= (A_1 \oplus B_1) \oplus C_0$			

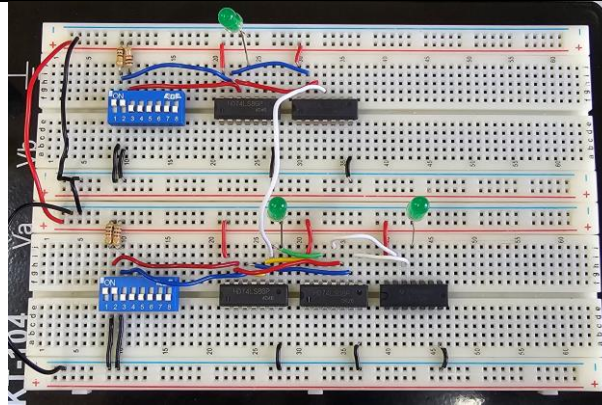
$$C_{out} = A_1C_0 + B_1C_0 + A_1B_1 = (A_1 \oplus B_1)C_0 + A_1B_1$$

Karnaugh map

C_0		S_0	
$A_0 \backslash B_0$	0 1	$A_0 \backslash B_0$	0 1
0	0 0	0	0 1
1	0 1	1	1 0
$A_0 \cdot B_0$		$A_0 \oplus B_0$	

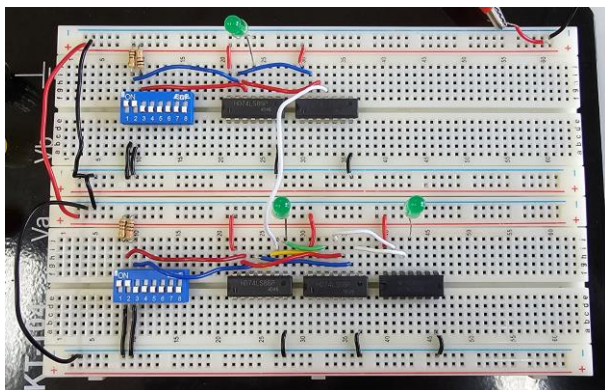
S_1		C_{out}	
$A_1 \backslash B_1 C_0$	00 01 11 10	$A_1 \backslash B_1 C_0$	00 01 11 10
0	0 1 0 1	0	0 0 1 0
1	1 0 1 0	1	0 0 1 1
$(A_1 \oplus B_1) \oplus C_0$		$A_1 B_1 + C_0(A_1 \oplus B_1)$	

Breadboard Wiring Photo

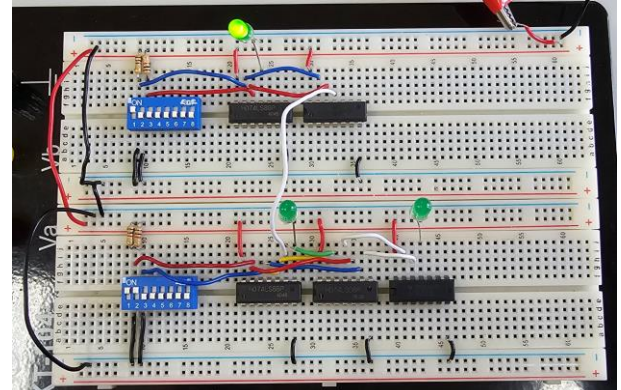


Breadboard Wiring Photos (The output must be connected to LEDs.)

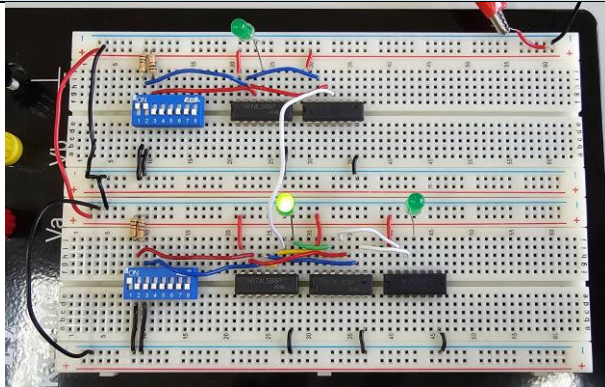
$A_1A_0 = 00, B_1B_0 = 00$



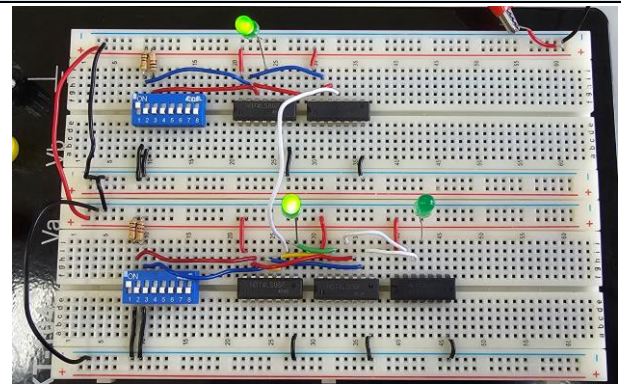
$A_1A_0 = 00, B_1B_0 = 01$



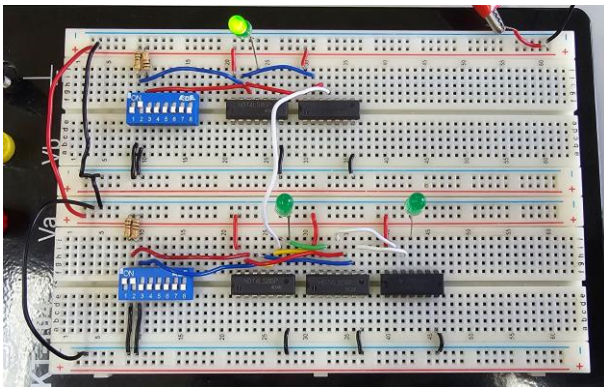
$A_1A_0 = 00, B_1B_0 = 10$



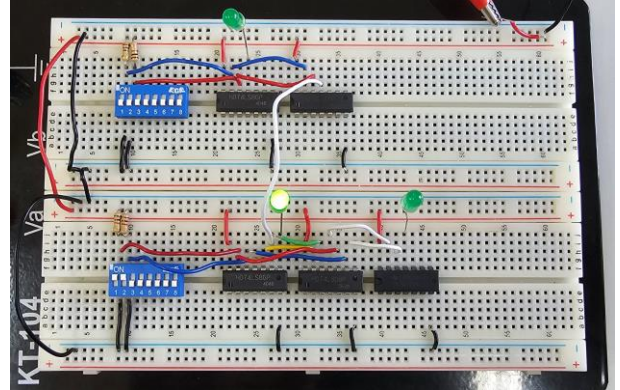
$A_1A_0 = 00, B_1B_0 = 11$



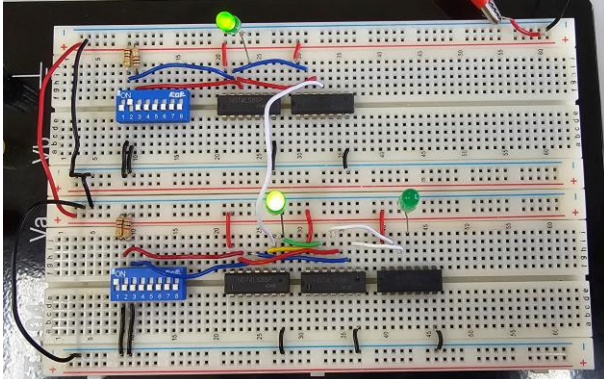
$$A_1A_0 = 01, B_1B_0 = 00$$



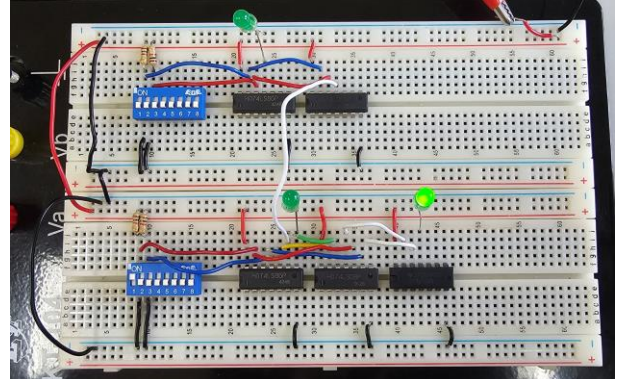
$$A_1A_0 = 01, B_1B_0 = 01$$



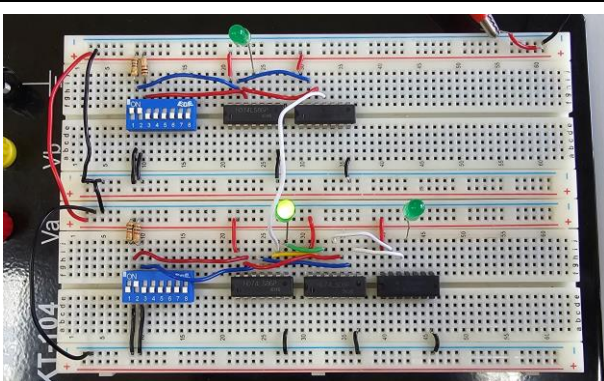
$$A_1A_0 = 01, B_1B_0 = 10$$



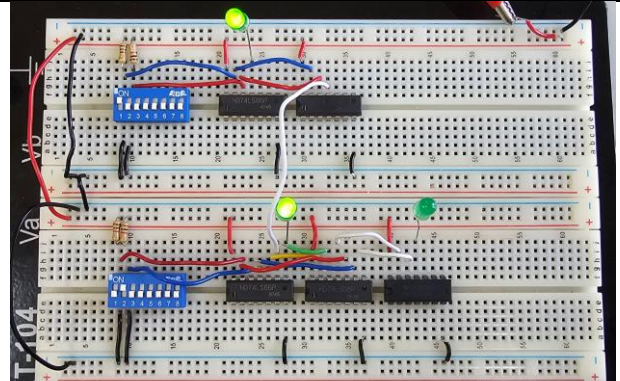
$$A_1A_0 = 01, B_1B_0 = 11$$



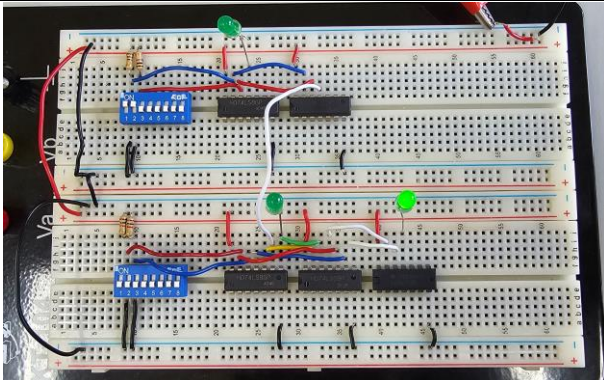
$$A_1A_0 = 10, B_1B_0 = 00$$



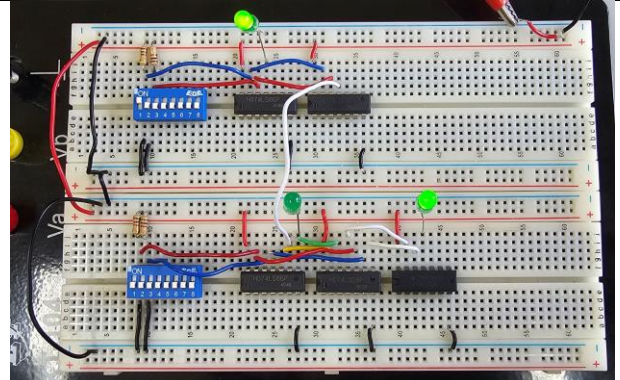
$$A_1A_0 = 10, B_1B_0 = 01$$



$$A_1A_0 = 10, B_1B_0 = 10$$

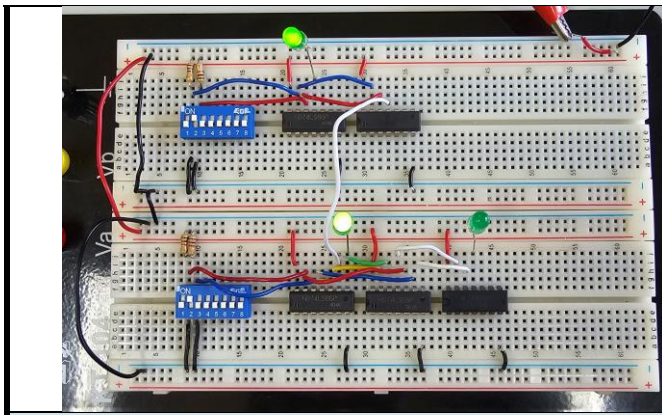


$$A_1A_0 = 10, B_1B_0 = 11$$

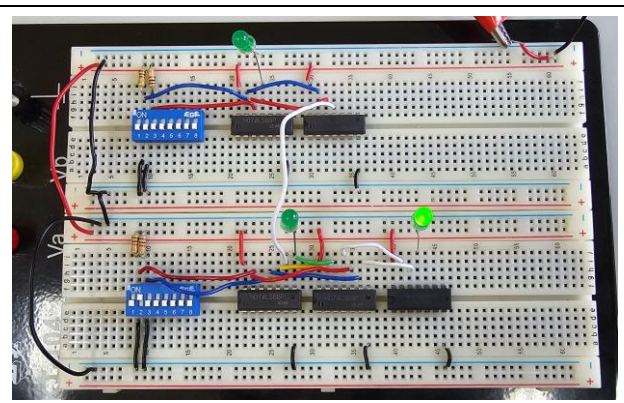


$$A_1A_0 = 11, B_1B_0 = 00$$

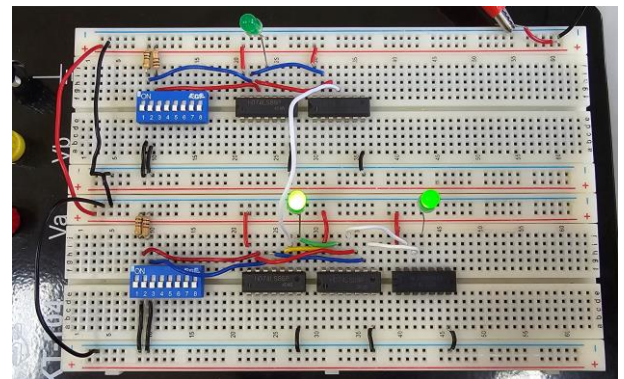
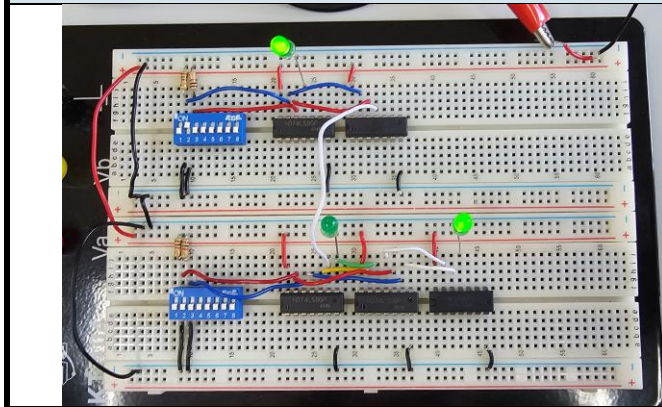
$$A_1A_0 = 11, B_1B_0 = 01$$



$A_1A_0 = 11, B_1B_0 = 10$



$A_1A_0 = 11, B_1B_0 = 11$



Lab Summary and Reflection :

Summary:

In this laboratory experiment, we implemented the half-adder and full-adder circuits as outlined in our Digital Logic course. The exercise involved constructing comprehensive truth tables and utilizing Karnaugh maps to simplify the Boolean expressions, ensuring clarity and precision in our design approach. Following the theoretical calculations, we translated our findings into practical applications by drawing detailed circuit diagrams and physically assembling the circuits on a breadboard with LEDs to visualize the output. This methodical process not only bridged the gap between theory and practice but also reinforced the understanding of how simple adders operate and how they form the building blocks for more complex arithmetic operations.

Reflection:

I found this lab session both challenging and engaging. Although the tasks demanded a high level of precision—from formulating the correct truth tables to simplifying Boolean expressions using the K-map method—the hands-on component was particularly rewarding. Having recently reviewed the first three chapters before the midterm, I felt confident in approaching the extended material, which allowed me to apply previously acquired knowledge effectively. The satisfaction of seeing the intended logical operations materialize through the successful operation of the breadboard circuitry, complete with LED outputs, underscored the importance of practical implementation in solidifying theoretical concepts. This experiment not only deepened my

understanding of digital logic design but also boosted my confidence in tackling progressively complex circuit designs in future assignments.